

1 Detecting causal models with the principle of
2 Mendelian randomization for genomic trios with
3 many confounding variables

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12 **Abstract**

13 The principle of Mendelian randomization (MR) makes it possible to infer the potential causal
14 relationship between two biological variables (e.g., expression of two genes, or between two
15 molecular phenotypes such as gene expression and DNA methylation), using a genetic variant
16 as an instrumental variable and thus forming a trio. Inference is often influenced by many other
17 variables (such as other genes and demographic variables). Being able to properly account for
18 confounding is therefore essential to causal inference. However, methods that can deal with
19 many confounding variables are limited. Here, we introduce a regression-based method under
20 this principle, namely MR for Genomic Networks in trios (MRGN), to detect diverse causal
21 models for genomic trios using individual-level data while accounting for many confounding
22 variables. We demonstrate the inference power and computational efficiency through exten-
23 sive simulation and application to examine how a single nucleotide polymorphism potentially
24 regulates its cis- and trans-target genes.

25 **1 Introduction**

26 An important goal in biology is to understand how genes regulate one another and how genetic
27 variation influences gene expression [1, 2, 3, 4]. Gene regulation involves complex networks of
28 high dimensions, which may be represented by directed networks where directed edges repre-
29 sent causal (or regulatory) relationships. Although the ultimate goal in biology is to reconstruct
30 gene regulatory networks at the genome level, it is often of interest to understand the causal re-
31 lationship between a molecular phenotype (e.g. the expression level) of two genes, or between
32 two molecular phenotypes (e.g., between gene expression and DNA methylation), which may
33 be influenced by many other variables such as other genes, other biological processes, demo-
34 graphic variables, etc.

35 To learn the (directed) causal relationship between two phenotypes, we can apply the prin-

36 ciple of Mendelian randomization (PMR) and leverage the information in genetic variants. The
 37 PMR assumes that genotypes in a natural population are randomly assigned, thus observational
 38 omics data can be viewed as being collected from a randomized experiment [5, 6]. This implies
 39 that an edge connecting a genotype to a phenotype will always point to the phenotype, but not
 40 the other way around [7, 8]. As a result, this constraint on edge direction makes it possible to
 41 break the symmetry of association and draw inferences about the regulatory structure among
 42 phenotypes [7, 9, 5].

43 Many methods have been developed under the PMR [10, 11, 12, 13, 7], although they often
 44 are interested in the mediation model $V \rightarrow T_1 \rightarrow T_2$, where V is the genetic variant, T_1 and T_2
 45 are the variables of interest. In this model, the mediator T_1 relays the regulatory information
 46 from V to the outcome T_2 ; for example, a genetic variant regulates the expression level of
 47 a transcription factor, and its reduced expression leads to its downstream target gene being
 48 under-expressed. However, other causal relationships may exist among the three variables.

49 Previously [7] developed MRPC, a method that goes beyond mediation and infers five
 50 basic causal models for a trio that are mutually exclusive (Figure 1) [7, 14]. They include a
 51 null model ($V \rightarrow T_1$ and no relationship between T_1 and T_2 , denoted M0) and four non-null
 52 models: mediation (denoted M1), v-structure (or reverse causation; $V \rightarrow T_1 \leftarrow T_2$, denoted
 53 M2), conditional independence (or horizontal pleiotropy; $T_1 \leftarrow V \rightarrow T_2$, denoted M3), and a
 54 fully connected model ($T_1 \leftarrow V \rightarrow T_2$ and $T_1 \leftrightarrow T_2$, denoted M4).

55 The true model behind M4 may be directed: the edge between T_1 and T_2 is $T_1 \rightarrow T_2$ or
 56 $T_1 \leftarrow T_2$. However, these two directions have the same likelihood and are therefore Markov
 57 equivalent. Without additional information, they are indistinguishable and only an undirected
 58 edge can be inferred.

59 The PMR and the five basic causal models require that no confounding variables can ex-
 60 plain away the directed edge between T_1 and T_2 , since this edge is our main interest. However,
 61 as mentioned earlier, many variables can influence the inference due to the complexity of gene

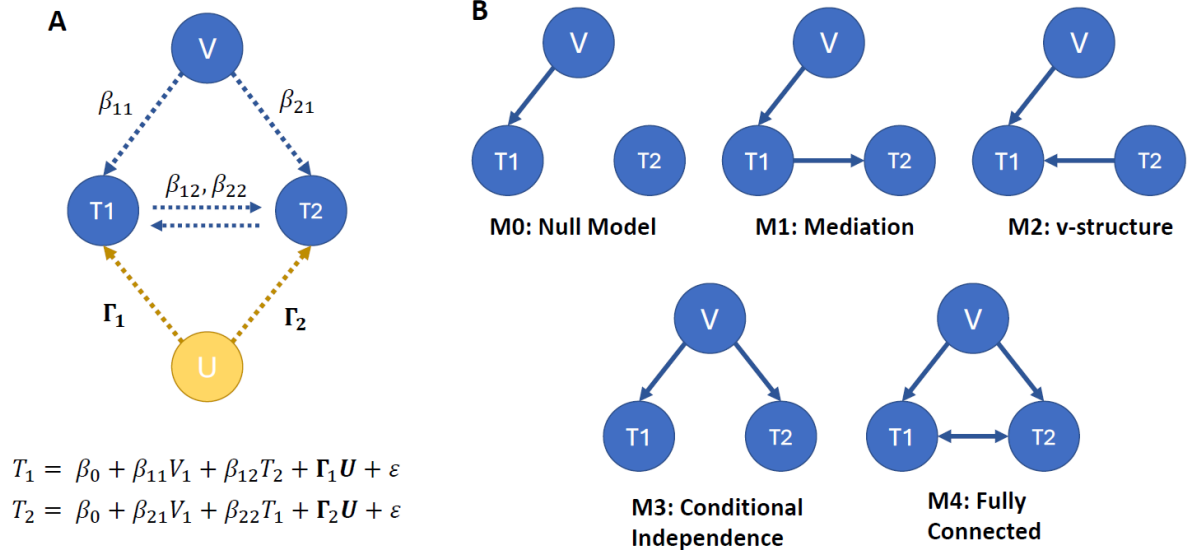


Figure 1: Causal models for a genetic trio. (A) A genomic trio includes a genetic variant (V), two molecular phenotypes (T_1 and T_2), and a set of confounders (U). (B) The five basic causal models for a trio that may be inferred under the principle of Mendelian randomization (adapted from [7]).

regulation. A confounding variable may influence the inference in different ways. [15] proposes three types of confounding for the $T_1 - T_2$ relationship: as a confounder, an intermediate variable, or a common child to T_1 and T_2 . Specifically, a confounder is a common parent to both T_1 and T_2 and can induce a false edge between T_1 and T_2 when unaccounted for. This is similar to T_1 and T_2 being marginally dependent in M3.

An intermediate variable acts like a mediator between T_1 and T_2 and can potentially explain away a true $T_1 \rightarrow T_2$ edge when this intermediate variable is accounted for in the inference, leading to a false negative conclusion. By contrast, a common child can induce a false edge between T_1 and T_2 when accounted for, leading to a false positive conclusion. This is because when the common child is accounted for in the inference, T_1 and T_2 become conditionally dependent, similar to their relationship in M2.

Accounting for confounding variables is therefore crucial in causal network inference [16, 15, 17, 18, 19]. Two basic approaches have emerged to address latent confounding between an exposure and outcome. In the first approach, candidate confounding variables are incorporated

into the graph as additional nodes in the network and also treated in the same way as other nodes; our MRPC method and several other methods take this approach [7, 18]. However, it is computationally intensive to include many confounding variables, as the computational complexity for graph inference grows super exponentially with the size of the graph. The second approach focuses on identifying and accounting for confounders [15, 17, 16]. This approach includes all the confounders as a whole in the conditional independence tests, and does not increase the size of the network. For example, we can include the confounders when testing the $T_1 - T_2$ edge in a trio. The trio remains a network of three nodes, but the tests are conditioned on these confounders. However, these methods focus on the mediation model.

Herein we present Mendelian Randomization for Genomic Networks (MRGN), a novel machine learning method for inferring the causal model of a trio from individual-level data in the presence of potentially hundreds of confounding variables. MRGN builds on MRPC and infers the five basic causal models. We take the second approach to confounding described above and focus on accounting for confounders, and not other types of confounding variables. This approach allows for the inclusion of many confounding variables all at once, while maintaining high power and fast computation times. We demonstrate the performance of MRGN and compare to other methods through extensive simulation studies. We further assess this method in our study of the cis- and trans-regulatory modes of genetic variants, using the whole blood data from the Genotype-Tissue Expression (GTEx) Consortium [20].

2 Methods

2.1 The MRGN model

The five causal models for a trio can each be described by a set of marginal and conditional dependence relations (Table 1), which may be encoded in the coefficients of two regression

models, assuming that T_1 and T_2 are centered so that the intercept is zero:

$$T_1 = \beta_{11}V + \beta_{12}T_2 + \mathbf{\Gamma}_1\mathbf{U} + \epsilon_1; \quad (1)$$

$$T_2 = \beta_{21}V + \beta_{22}T_1 + \mathbf{\Gamma}_2\mathbf{U} + \epsilon_2, \quad (2)$$

where β s are coefficients of interest, vector $\mathbf{\Gamma}_i$ the coefficients for the confounder vector $\mathbf{U}_i$, and ϵ_i the random error that follows a normal distribution with mean 0 and standard deviation σ_i , where $i = 1, 2$.

Our conditional independence tests test whether each β is zero, as each β represents the association given the presence of other variables. For example, in the mediation model of $V \rightarrow T_1 \rightarrow T_2$, any two variables are dependent on each other, but T_2 and V are conditionally independent given T_1 . These relations translate to (i) β_{21} being zero, which represents the association between V and T_2 conditioned on T_1 and the confounders $\mathbf{U}$, and (ii) the other three β s being nonzero, which represent those associations not being explained away by other variables. Similarly, models M0 and M3 are uniquely determined by conditional associations, for which we only need to conduct conditional independence tests.

In the v-structure model M2, consider the configuration $V \rightarrow T_1 \leftarrow T_2$. As the parents of T_1 , nodes V and T_2 are marginally independent but conditionally dependent given the presence of their child T_1 . Therefore, all the β s, which represent conditional associations, should be nonzero. Similarly, all the β s are also nonzero in the fully connected model M4. The distinction between M2 and M4 lies in the marginal associations: V and T_2 are marginally independent in M2 but are marginally dependent in M4. We therefore perform additional tests on marginal dependence to distinguish these two models.

We perform the standard t tests on the regression coefficients for conditional dependence and the standard test on Pearson correlation for marginal dependence. When the data is not informative enough for any of the basic models to be inferred, we categorize the trio as “other”.

Table 1: Conditional and marginal independence tests in MRGN. Regression coefficients β_{11} and β_{12} (from Equation 1) and β_{21} and β_{22} (from Equation 2) are used to test conditional independence. Marginal independence can be tested through correlations r_{V,T_1} and r_{V,T_2} . The first three models each have two possible configurations.

		Conditional Test				Marginal Test	
Null Hypothesis		$T_1 \perp\!\!\!\perp V$	$T_1 \perp\!\!\!\perp T_2$	$T_2 \perp\!\!\!\perp V$	$T_2 \perp\!\!\!\perp T_1$	$V \perp\!\!\!\perp T_1$	$V \perp\!\!\!\perp T_2$
Conditioned on		$T_2, \mathbf{U}$	$V, \mathbf{U}$	$T_1, \mathbf{U}$	$V, \mathbf{U}$	–	–
Parameter to Test		β_{11}	β_{12}	β_{21}	β_{22}	r_{V,T_1}	r_{V,T_2}
Model	Configuration						
M_0	$V \rightarrow T_1; T_2$ is singleton	$\neq 0$	$= 0$	$= 0$	$= 0$	–	–
	$V \rightarrow T_2; T_1$ is singleton	$= 0$	$= 0$	$\neq 0$	$= 0$	–	–
M_1	$V \rightarrow T_1 \rightarrow T_2$	$\neq 0$	$\neq 0$	$= 0$	$\neq 0$	–	–
	$V \rightarrow T_2 \rightarrow T_1$	$= 0$	$\neq 0$	$\neq 0$	$\neq 0$	–	–
M_2	$V \rightarrow T_1 \leftarrow T_2$	$\neq 0$	$\neq 0$	$\neq 0$	$\neq 0$	$\neq 0$	$= 0$
	$V \rightarrow T_2 \leftarrow T_1$	$\neq 0$	$\neq 0$	$\neq 0$	$\neq 0$	$= 0$	$\neq 0$
M_3	$T_1 \leftarrow V \rightarrow T_2$	$\neq 0$	$= 0$	$\neq 0$	$= 0$	–	–
M_4	$T_1 \leftarrow V \rightarrow T_2;$	$\neq 0$	$\neq 0$	$\neq 0$	$\neq 0$	$\neq 0$	$\neq 0$
	$T_1 \longleftrightarrow T_2$						

2.2 Confounder selection

We perform confounder selection before regression, using a two-step procedure similar to that described in [15]. First, we identify candidate confounders from a pool of potential confounding variables. Confounders, which are defined as parents of T_1 and T_2 , should be independent of V . We therefore exclude variables that are marginally associated with V . Next, we select

126 confounders based on their statistical association with either T_1 or T_2 . Similar to [15], we test
 127 whether the regression of both T_1 and T_2 on a candidate confounder is statistically significant.
 128 Both steps employ the q -value method to control the false discovery rate (FDR) [21].

129 In our R implementation, we substantially improve computational efficiency in two key
 130 ways. First, we implemented parallelization in both steps, allowing for faster evaluation when
 131 the pool of confounders is large. Second, we used the R package *propagate* for correlation
 132 matrix computation by partitioning the problem into smaller, memory-efficient blocks [22],
 133 which scales more efficiently to large genomic datasets.

134 2.3 Simulation study

135 Simulation setup

136 Each simulated trio consists of a bi-allelic genetic variant V with three genotypes and two
 137 genes (T_1, T_2) connected under one of the five causal models. Nodes in a trio are generated in
 138 topological order. The genotype V_i is simulated first under Hardy–Weinberg equilibrium:

$$\Pr(V_i = 0) = (1 - \theta)^2, \quad \Pr(V_i = 1) = 2\theta(1 - \theta), \quad \Pr(V_i = 2) = \theta^2,$$

139 where the minor allele frequency θ is sampled uniformly from (0.01, 0.5), such that roughly
 140 10% of trios contain a less common allele with a $< 5\%$ frequency.

141 Each remaining node Y_j (a gene or a confounder) is sampled from

$$Y_j \sim N\left(\sum_{i \in S_j} \beta_i Y_i, \sigma^2\right),$$

142 where S_j is the parent set of Y_j , which may include V , the other gene, or confounders. The
 143 effect size of V on its target genes is sampled uniformly from [0.5, 1.5] [15]. The effect size
 144 between T_1 and T_2 is sampled uniformly from [0.5, 1]. Residual variance is set to $\sigma = c\beta$,

145 where β is the T_1 - T_2 effect and c sampled uniformly from $[0.3, 1.5]$.

146 For **M4**, the true edge between T_1 and T_2 can be in either direction. For simplicity, we fix
147 all M4 trios to the configuration $T_1 \leftarrow V \rightarrow T_2$ with $T_1 \rightarrow T_2$.

148 We consider three types of confounding variables in simulation: confounders ($T_1 \leftarrow C \rightarrow$
149 T_2 , where C is a confounding variable), intermediate variables ($T_1 \rightarrow C \rightarrow T_2$), and common
150 children ($T_1 \rightarrow C \leftarrow T_2$) (see Supplementary Figure 1) [15]. For each trio, we include a vary-
151 ing number of confounders (between 1 and 50), one intermediate variable, and one common
152 child. Effect sizes for these variables are sampled independently: confounder and intermediate
153 effects from $[0.15, 0.5]$ and common-child effects from $[0.5, 1.5]$, each with a 50% probability
154 of being negative.

155 We simulate 1,500 trios by sampling from the five causal models with equal probability,
156 and obtain approximately 300 trios per model. For each trio, we draw a distinct combination
157 of effect sizes, noise level, and minor allele frequency. The sample size is fixed at $n = 670$,
158 matching the whole-blood sample size of GTEx v8 [20] in our real-data application.

159 **Performance metrics**

160 We evaluated three sets of recall and precision to quantify the performance on the entire model
161 of a trio, on individual edges and specifically on the $T_1 - T_2$ edge.

- 162 1. On the causal model type: a trio is considered correctly inferred when its inferred model
163 type matches its true type, thus requiring all edges to be inferred correctly.

$$\text{Recall}_{\text{model M}} = \frac{\text{Number of correctly inferred trios}}{\text{Total number of trios of the causal model M}}; \quad (3)$$

$$\text{Precision}_{\text{model M}} = \frac{\text{Number of correctly inferred trios}}{\text{Total number of trios inferred to be of the causal model M}}. \quad (4)$$

- 164 2. On individual edges: we first calculated recall and precision based on individual edges

165 in a single trio. Similar to our earlier work [7], we counted an edge with the wrong
 166 direction as a half correct edge.

$$\text{Recall}_{\text{edges}} = \frac{\text{Number of correct edges in inferred graph}}{\text{Total edges in true graph}}; \quad (5)$$

$$\text{Precision}_{\text{edges}} = \frac{\text{Number of correct edges In inferred graph}}{\text{Total edges in inferred graph}}. \quad (6)$$

167 We then calculated the mean metric across all trios.

168 3. On the $T_1 - T_2$ edge (allowing for comparison with the GMAC method): the direct of
 169 this edge was ignored for this evaluation. Across all trios,

$$\text{Recall}_{T_1 - T_2} = \frac{\text{Number of trios with a correctly identified } T_1 - T_2 \text{ edge}}{\text{Total trios with a true } T_1 - T_2 \text{ edge}}; \quad (7)$$

$$\text{Precision}_{T_1 - T_2} = \frac{\text{Number of trios with a correctly identified } T_1 - T_2 \text{ edge}}{\text{Total trios with an inferred } T_1 - T_2 \text{ edge}}. \quad (8)$$

170 3 Results

171 3.1 Simulation Results

172 MRGN improves power and precision in the presence of confounding

173 We applied MRGN with two types of corrections for multiple testing for confounder selection:
 174 controlling the FDR at 5% with the q -value method (denote “CS- q ”), and controlling the type I
 175 error rate at $\alpha < 0.01$ (denoted “CS- α ”). These two corrections produced qualitatively similar
 176 results in recall and precision on the inference of the trio (Table 2). Both versions achieve sub-
 177 stantially higher recall for the mediation, v-structure, and fully connected models than MRPC
 178 does. All these models contain a $T_1 - T_2$ edge, indicating the sensitivity of MRGN in detecting
 179 the $T_1 - T_2$ edge. On the other hand, this sensitivity leads to reduced power to detect the null

180 and conditional independence models, which have no $T_1 - T_2$ edge. Remarkably, MRGN also
 181 achieves equivalent or better precision for nearly all models.

Table 2: Comparison of MRGN, MRPC and GMAC on simulated trios with confounders.

The oracle performance of MRGN using true confounders is shown to illustrate the upper bound due to noise in the data. For the other columns, the highest precision and recall in each row is highlighted. CS-q: confounder selection with the q -value method for FDR control at 5%. CS- α : confounder selection with the type I error rate set at $\alpha < 0.01$ and no FDR control. MRPC cannot handle too many variables in reasonable runtime; we therefore did not run MRPC with the many confounders identified from CS- α . Rows 1-5 show model type-based performance, row 6 for all three edges in a trio, and the last row for only the $T_1 - T_2$ edge. * indicates that similar values are produced by the GMAC method.

	MRGN (True Confs.)		MRGN (CS-q)		MRGN (CS- α)		MRPC (CS-q)	
	Recall	Prec.	Recall	Prec.	Recall	Prec.	Recall	Prec.
Null Model (M0)	0.562	0.995	0.452	1.000	0.477	0.951	0.836	0.532
Mediation (M1)	0.982	0.516	0.877	0.438	0.883	0.429	0.097	0.147
V-structure (M2)	0.919	0.985	0.892	0.925	0.842	0.968	0.377	0.443
Cond. Indep. (M3)	0.515	0.989	0.283	0.910	0.305	0.872	0.609	0.828
Fully Connect. (M4)	0.600	0.566	0.501	0.425	0.464	0.419	0.217	0.592
All Edges	0.945	0.893	0.931	0.864	0.911	0.857	0.770	0.892
$T_1 - T_2$	1.000	0.767	0.994*	0.703*	0.9787	0.7071	0.7884	0.916

182 At the edge level, MRGN recovers more true edges than MRPC does, while maintaining a
 183 high degree of precision. For the $T_1 - T_2$ edge, MRGN matches the performance of GMAC
 184 (Supplementary Figure 2), as the GMAC test for mediation is a special case of our method. The
 185 similar performance on the $T_1 - T_2$ edge further shows that the notion of “mediation” detected
 186 by GMAC is broad and an inferred mediation edge by GMAC may correspond to different
 187 underlying causal models.

MRGN handles many confounding variables

MRGNTrio and GMAC performed similarly well in recall and maintained consistently high recall for the $T_1 - T_2$ edge, even under substantial confounding, with precision remaining relatively unchanged (Supplementary Figure 2). Additionally, recall and precision across all edges remained high, indicating that MRGNTrio can accurately infer most trio edges, except in cases with many (> 35) confounders, where recall declined but still exceeded that of MRPC (Supplementary Figure 3).

The impact of confounder selection on causal network inference

Accurate selection of confounding variables is a critical step preceding causal inference in Mendelian randomization studies, as errors at this stage can propagate to the causal model by omitting informative genetic or environmental covariates. In our simulation study, we compared two strategies for selecting confounders from a pool of candidate covariates: CSnoFDR, which selects confounders without false discovery rate (FDR) correction, and CSFDR, which controls the FDR at 5%. We first evaluated both methods based on their ability to identify the true confounders, measured by precision and recall. Overall, both methods achieved recall greater than 90% when the number of true confounders per trio was modest (e.g., fewer than 25; see Figure 2). However, CSFDR showed reduced performance when for trios with many true confounders, with average recall to detect true confounders dropping to approximately 65%. Conversely, CSnoFDR achieved nearly perfect recall across all simulations, but at the expense of very low precision, indicating that it selected many false confounders in addition to the true ones.

When we examined how confounder selection affected causal inference performance, we found no clear relationship between the number of false-positive confounders and MRGNTrio's ability to accurately infer causal edges (Figure 3). In contrast, falsely omitted confounders had a strong negative impact on causal inference performance. Overall, we observed little differ-

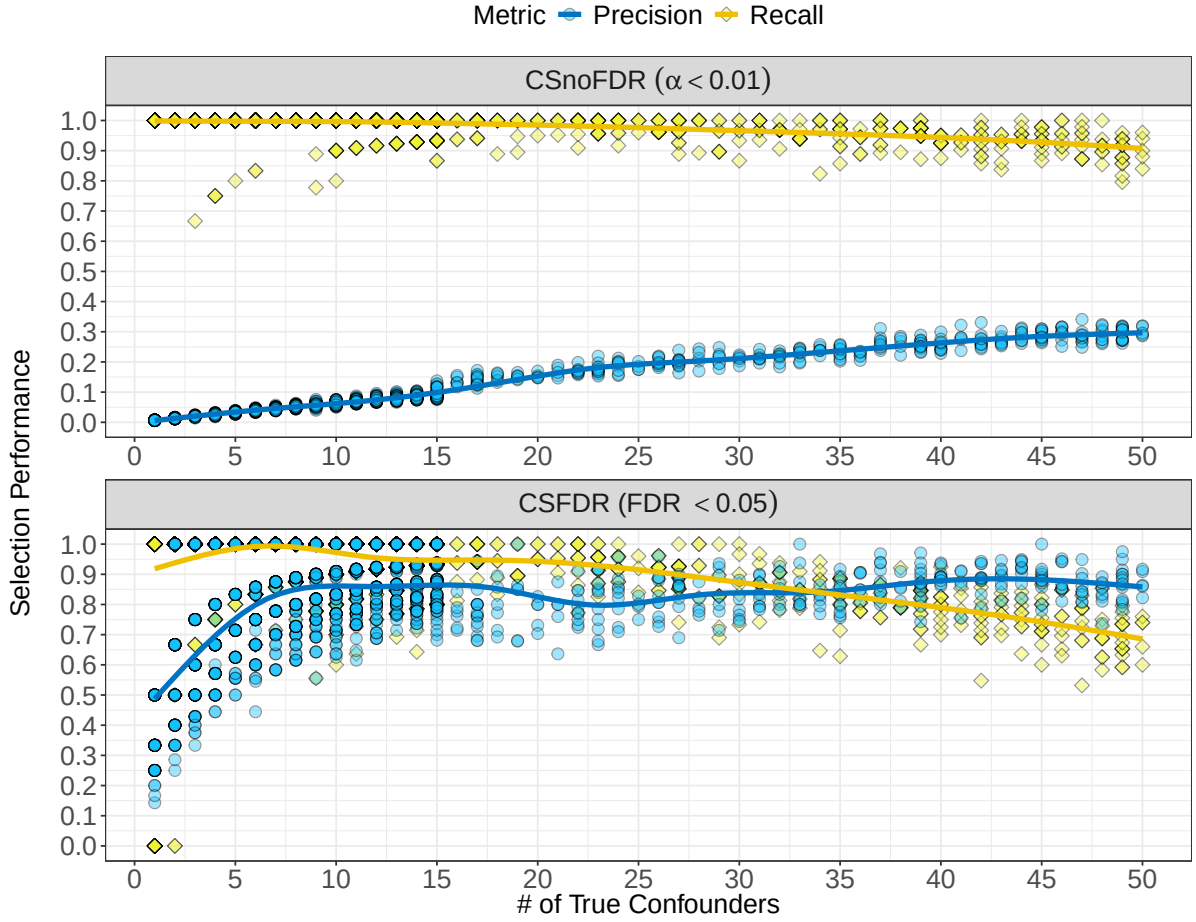


Figure 2: Precision–recall curves comparing confounder selection performance of the CSFDR and CSnoFDR methods. Recall is defined as the proportion of true confounders correctly identified and selected for the inference model while precision is the proportion of true confounders selected out of all confounders selected.

213 ence in causal inference performance between the two methods in our simulation study. This
 214 may be because only a small number of trios were simulated with a large number of confound-
 215 ing variables (e.g., more than 35), limiting our ability to detect any differences between the two
 216 methods. Although CSnoFDR is more liberal, it may be preferable in scenarios where many
 217 potential covariates are expected for a trio.

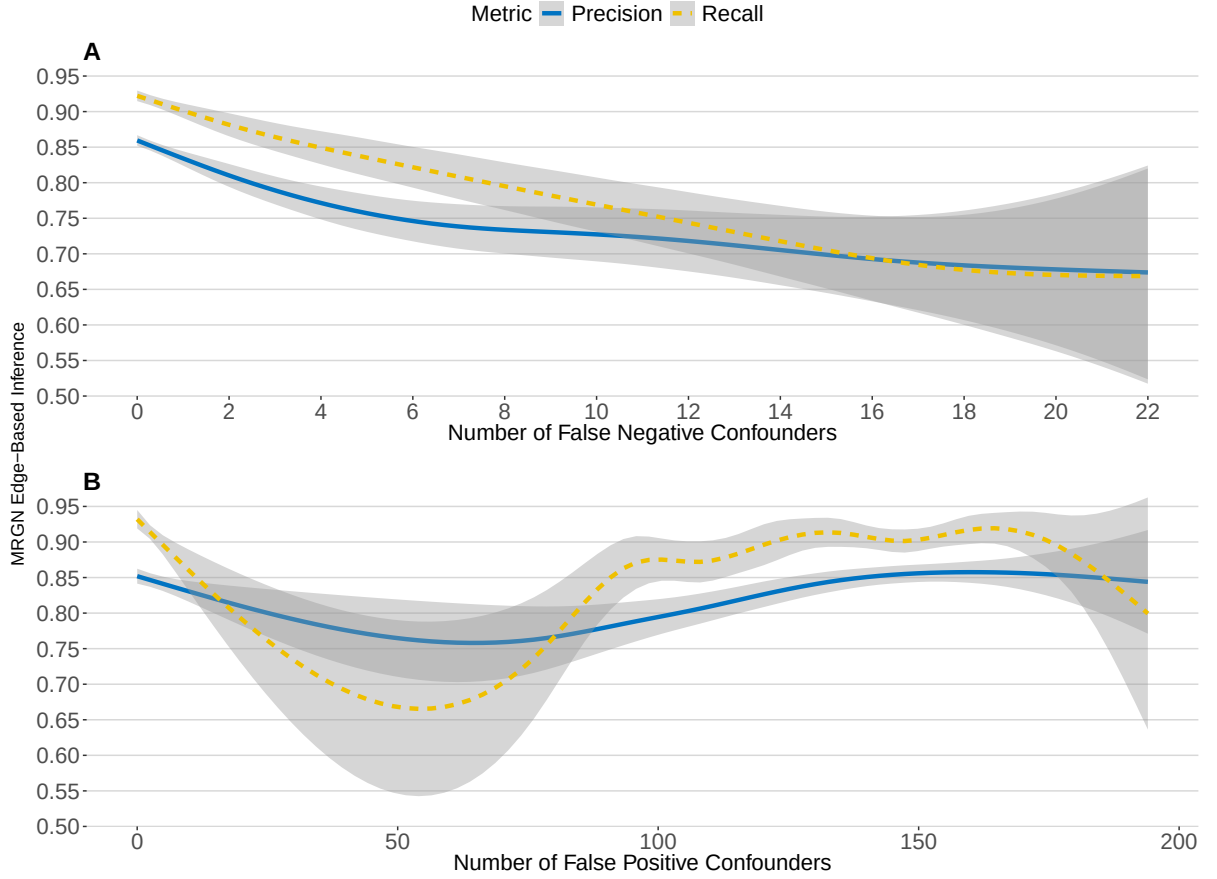


Figure 3: Relationship between confounder selection and causal inference performance measured as precision and recall of edge prediction. (A) Edge-based performance of MRGN_{Trio} on simulated trios as a function of the number of true confounders falsely omitted from the analysis. (B) Edge-based performance of MRGN_{Trio} on simulated trios as a function of the number of falsely included confounders.

Runtime comparison

We compare our MRGN method to two related PMR-based causal inference methods: Genomic Mediation analysis with Adaptive Confounding adjustment (GMAC) [15] and Mendelian Randomization PC (MRPC) [7]. Unlike MRGN and MRPC, GMAC is a mediation - based method that is restricted to inferring only the causal edge between T_1 and T_2 . To that end, we only compare MRGN and MRPC to GMAC on the performance for predicting the $T_1 - T_2$ edge. We set the number of permutations to 1000 and the test threshold for all marginal and conditional tests to be $\alpha = 0.01$ which is approximately a Bonferroni correction. For MRPC, we use the ADDIS FDR control, Gaussian CI test, and set the FDR to 5%. For MRGN, we employ the permutation test whenever the minor allele is observed in less than 5% of samples. GMAC, however, applies the permutation test described in to all trios regardless of allele frequency. For both methods, we set the number of permutations to 1000.

When analyzing these trios, we assumed that all the confounding variables were available but we did not know which confounding variables were associated with which trio. In our inference, we first performed the confounding variable selection method of [15] to all the confounding variables, before applying three methods: MRGN, GMAC and MRPC to each trio with the identified confounders. For all these methods, we used the q -value [21, 24] method for FDR control in confounder selection (denoted CSFDR). For MRGN, we also performed confounder selection without the FDR control (denoted CSnoFDR) (see “Application of causal inference methods”). We further simulated another set of 300 trios with 15–50 confounders (median: 34), representing a scenario with moderate to extreme genetic confounding. The candidate confounder pool trios in this scenario contained about 11,000 variables.

Following the approach of [25], we run GMAC twice on all trios, first with T_1 as the mediator and then with T_2 as the “mediator” and take significant mediation at the $\alpha < 0.05$ and $\alpha < 0.01$ levels. To give GMAC the best chance at predicting the $T_1 - T_2$ edge across all causal models, we combine the inference results from the tests with T_1 as the mediator

with the tests with T_2 as a mediator. Combining the inference from using each gene as the mediator establishes four possible results: positive with T_1 as the mediator, positive with T_2 as the mediator, both positive, or both negative. To afford GMAC greater power across the five models, we take any of the aforementioned positive cases for mediation as an indication of the existence of the $T_1 - T_2$ edge.

To emulate a real statistical analysis, we include confounding variable selection as a preliminary step to inference. The confounder selection method of [15] requires the input of a pool of potential confounding variables. In simulation, we construct this pool of potential confounding variables by aggregating the true confounding variables from all simulated trios together (i.e. all confounders, intermediate and common child variables). We control the false discovery rate (FDR) for filtering common child and intermediate variables from the candidate pool at the 10% level. Furthermore we consider two settings for selecting confounders from the pool post-filtering (i) selecting confounders with no FDR correction and significant confounders selected at the $\alpha < 0.01$ (denoted CSnoFDR) and (ii) using the qvalue method to control the FDR of selected confounders at the 5% level (denoted CSplusFDR). The latter setting is the default for selecting confounders using the GMAC software.

Our analysis involves two major steps: confounder selection and causal network inference. We report the computation time for each step separately. We conducted a comparative analysis of the time required for confounder selection across varying sizes of the confounder covariate pool (100, 500, or 1000 covariates) and different numbers of trios for which confounders needed to be identified (500 or 1000), resulting in a total of six replicates. Notably, in the most intensive case 1000 covariates and 1000 trios, our method accomplished confounder selection within 16 minutes (Supplementary Figure 4). This represents a notable gain in computation efficiency compared to the implementation provided by GMAC, which took approximately 28 minutes. Although our confounder selection procedure is similar to that of [15], our implementation is more efficient due to the use of the R package propagate [22], which enables

fast correlation calculations between each trio’s target genes and the candidate covariate pool, significantly reducing the time required for confounder selection.

In causal network inference, MRGN is substantially faster than both MRPC and GMAC (Table 3). The median time to analyze a trio using the same set of confounders was 0.0065 seconds for MRGN, compared to 0.3557 seconds for MRPC and 0.7088 seconds for GMAC. This makes MRGN over 50 times faster than MRPC and more than 100 times faster than GMAC, on average.

The extended inference time for MRPC is largely due to the ADDIS method used for false discovery rate (FDR) control, which was particularly slow for some trios. When MRPC is applied using the more conservative LOND method for FDR control, its runtime becomes comparable to that of MRGN. For GMAC, we note that our comparison used the default single-threaded implementation provided in its R package. Enabling the parallelization option available in GMAC can substantially reduce its computational time.

Table 3: Inference times for MRGN, MRPC-ADDIS, and GMAC on simulated trios. MRGN shows a substantially faster median inference time per trio compared to both MRPC-ADDIS and GMAC, demonstrating its computational efficiency in the presence of confounding.

Method	Median time (s)	IQR (s)
MRGNTrio	0.0105	0.0052
MRPC-ADDIS	0.4601	9.0117
GMAC	1.5373	1.4831

Confounding contributes to an inflated type I error rate

In our simulation study, we observed significantly higher rates of false positive mediation for the GMAC method than those reported by [15]. In their simulations under the null model, [15] reported type I error rates (false mediation) consistent with the significance level used

287 to test for mediation. However, in our simulations, we observed type I error rates for the
288 $T_1 - T_2$ edge under the null model to be as high as 75% for both MRGN and GMAC at the
289 $\alpha < 0.01$ significance cutoff. We attribute these differences to several distinctions between our
290 simulation framework and theirs.

291 First, we simulated a substantially larger number of confounders for each trio (1–50 across
292 all trios in our simulations, compared to 1–2 in [15]). Additionally, to construct the pool of
293 confounders used for selection, we aggregated all confounding variables from all simulated
294 trios, resulting in a large candidate pool of more than 10,000 variables. We also simulated
295 each trio to have one common child and one intermediate variable, which were included in the
296 candidate covariate pool. In contrast, [15] simulated a much smaller pool of 350 confounders,
297 yielding an average of approximately two confounders per trio, with many trios having no
298 confounders. Furthermore, their simulations treated each type of variable (confounder, com-
299 mon child, intermediate variable) separately, whereas our simulations included all three types
300 simultaneously.

301 To validate the error rates reported by [15], we incorporated their simulation design into
302 our framework to examine the effect of confounders on type I error (false mediation) under
303 the null model. We simulated 100 additional trios for each of seven scenarios containing 2,
304 4, 6, 8, 10, 12, or 15 confounders, excluding common child and intermediate variables, and
305 applied GMAC with CSFDR confounder selection using our approach of aggregating covari-
306 ates across simulated trios to construct the confounding variable pool. The scenario with two
307 confounders is comparable to the null-model simulations reported by [15], while the remaining
308 scenarios align with our simulation settings. We successfully replicated the type I error rates
309 reported by [15] in the two-confounder scenario (Supplementary Figure 9), whereas all other
310 scenarios showed increasingly inflated rates of false mediation that scaled with the number of
311 confounders and the size of the covariate pool. These results demonstrate that substantial con-
312 founding leads to markedly inflated false positive mediation and further support the low power

313 to detect the null and conditional independence models observed in our simulation study.

314 **3.2 Application to GTEx data**

315 The GTEx consortium has mapped a large number of eQTLs and their associated cis target
316 genes (cis-eQTLs and cis-target genes) for most genes in the human genome across many hu-
317 man tissue types [20]. Recent studies have focused on identifying cis-target genes that mediate
318 the relationship between the eQTL and its associated trans-target genes [15]. We previously
319 have applied the MRPC method to all the GTEx tissues to detect diverse causal models for
320 such trios: an eQTL and its associated cis- and trans-target genes, where the genes are protein
321 coding genes or long non-coding RNAs (lncRNAs) [25].

322 We implement the preprocessing and normalization pipeline from Section 2.2 to construct
323 trios for WholeBlood tissue from version 8 of GTEx. Each trio consists of an eQTL and its
324 highest association “cis” and “trans” target genes. We define cis genes as genes less than one
325 megabase from the eQTL and trans genes as genes residing more than one megabase from the
326 eQTL or on a different chromosome. As noted in Section 2.2 individual trios may share the
327 same eQTL or the same gene, and the gene may be a cis-gene in one trio but a trans gene in
328 another trio.

329 In GTEx version 8, the number of available samples for whole blood is 670. After prepro-
330 cessing, 8,823 trios were identified. To eliminate bias induced by mismapping transcripts, we
331 restricted our final set of trios to protein coding genes or long non-coding RNA genes [15].
332 This additional filtering step reduces the number of trios to 3,248. Many of eQTLs provided by
333 the GTEx consortium contain missing values. Since the GMAC algorithm is unable to handle
334 missing values, we used multiple correspondence analysis implemented in the *missMDA* pack-
335 age in *R* [26] to impute the missing genotypes of individuals using the collection of unique
336 eQTLs from all GTEx trios analyzed for whole blood.

337 **Confounding variable selection and filtering for GTEx trios**

338 We performed principal component analysis (PCA) on the PEER-normalized, genome-wide
339 gene expression data and used the resulting principal component (PC) scores as the pool of
340 candidate covariates for confounder selection. Each PC summarizes the regulatory effects of a
341 large number of genes across the genome. Because each trio represents a small component of
342 a larger regulatory network, PCs that correlate with a trio may capture potential confounding
343 effects arising from pleiotropy. We performed confounder selection as described in Confound-
344 ing Variable Selection and applied common-child and intermediate-variable filtering to obtain
345 a final set of confounders for each trio. Following our findings from simulation, we applied
346 MRGNTrio using the CSnoFDR setting and selected confounders at a significance threshold of
347 $\alpha < 0.05$. While the simulation study used a more stringent cutoff of $\alpha < 0.01$, we relaxed this
348 threshold here because the more stringent cutoff produced nearly the same set of confounders
349 as the CSFDR setting when applied to the whole-blood PC score matrix. For each trio, we also
350 included sex, platform, and PCR batch as additional covariates.

351 **Results**

352 We applied MRGNTrio combined with the CSnoFDR confounder selection procedure to 3,248
353 eQTL trios for whole blood in GTEx. We compared the results from MRGNTrio with those
354 from MRPC (using the ADDIS method to control the overall FDR of conditional tests) and
355 GMAC [7, 27]. Principal component analysis (PCA) was performed across all genes in whole
356 blood to derive 670 principal components, which served as the pool of candidate confounding
357 variables. For each trio, we identified a median of 51 confounders (SD = 6.38) (Figure 4).

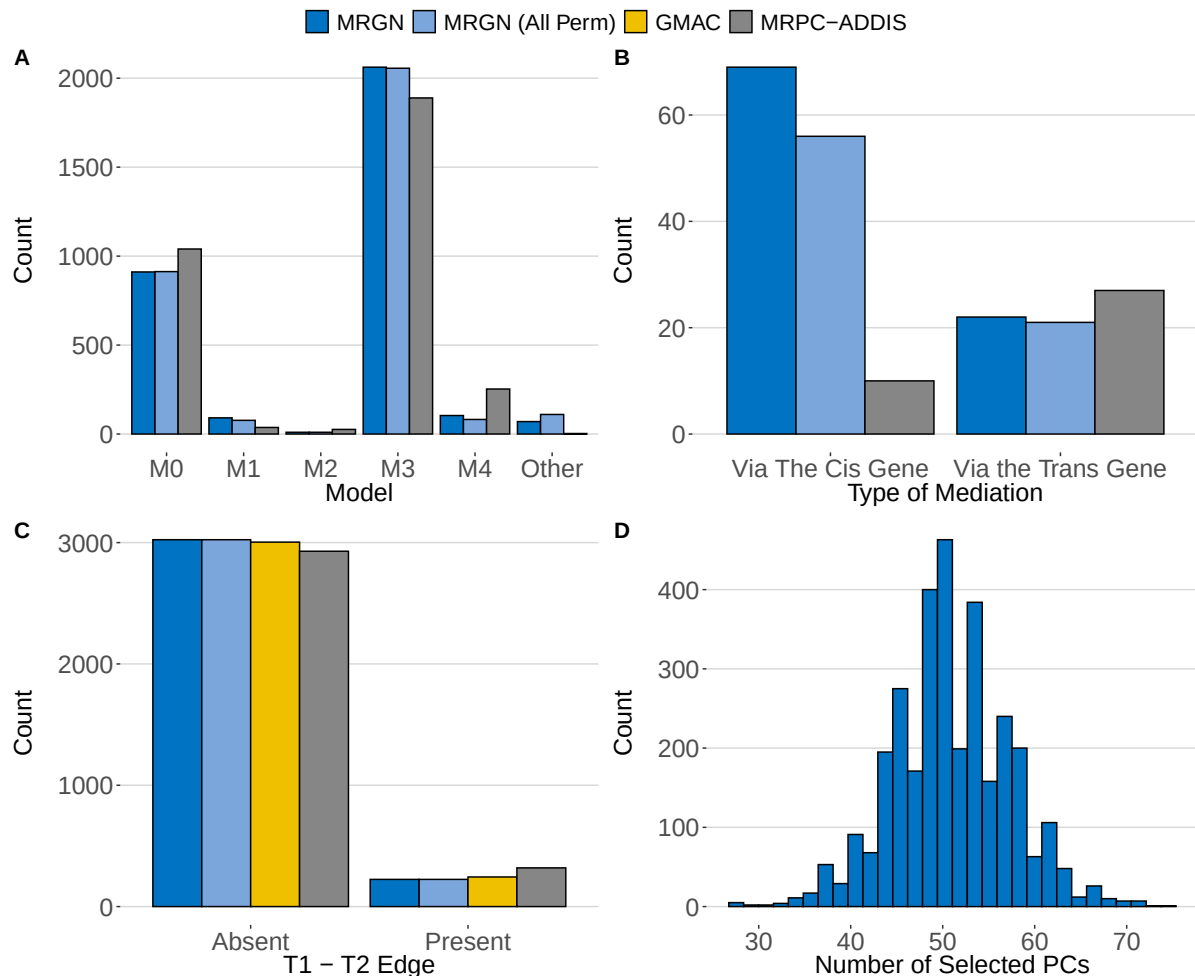


Figure 4: Comparative analysis of MRGN, MRPC [28] and GMAC [15] on GTEx human whole blood trios. We applied MRGN, GMAC, and MRPC (with ADDIS for FDR control) to genomic trios. (A) Count of inferred causal models (M1–M4 and Other) comparing MRGN and MRPC. (B) Comparison of specific mediation pathways (via cis-gene vs. trans-gene) identified by MRGN and GMAC. (C) Total counts of trios where the $T_1 - T_2$ causal edge was inferred as Present or Absent across all four methods[cite: 33, 34, 36]. (D) Histogram displaying the frequency distribution of the number of Principal Components (PCs) selected as confounders by the CSnoFDR method.

After applying MRGN_{Trio}, 911 trios were classified as the null model, 91 as mediation, 10 as v-structures, 2,062 as conditionally independent, and 104 as fully connected. The remaining 70 trios were categorized as “other.” We compared these results with our previous work using MRPC [25], and found that MRGN_{Trio} identified more than twice as many cases of causal me-

362 diation and 173 additional cases of conditional independence. Similar to MRPC, MRGNTrio
 363 also distinguishes two types of mediation: (1) cis-gene mediation $V \rightarrow T_1 \rightarrow T_2$, where the
 364 cis-gene mediates the signal from genetic variant to the trans-gene, and (2) trans-gene media-
 365 tion $V \rightarrow T_2 \rightarrow T_1$, where the trans-gene acts as the mediator between the genetic variant and
 366 the cis-gene. Among the 91 trios identified as mediation by MRGNTrio, 69 (75.5%) had the
 367 cis-gene as the mediator and 22 (24.5%) had the trans-gene as the mediator (Supplementary
 368 Table 3). Compared to MRPC, MRGNTrio identified not only more mediation trios overall,
 369 but also a higher proportion of cis-gene mediation. However, only 8 cis-gene mediation trios
 370 and 6 trans-gene mediation trios were identified by both methods.

371 We further investigated trios where MRGNTrio inferred mediation but MRPC did not and
 372 found that these discrepancies were primarily driven by differences in the sets of selected con-
 373 founders (see Supplementary Text 1; Examples 1–2). In two illustrative cases, MRGNTrio
 374 selected 40 and 43 confounders, whereas MRPC selected only three and none. When we reran
 375 MRGNTrio using the same confounder set and inference model as MRPC, both methods pro-
 376 duced identical results, confirming that accurate identification of confounders is essential for
 377 consistent causal inference.

378 **MRGN is robust to model misspecification in causal mediation**

379 Based on our simulation and application findings, we conducted a secondary analysis to in-
 380 vestigate how model misspecification affects the detection of causal mediation. In particular,
 381 we examined how inaccuracies in the inference model, particularly in terms of confounding
 382 variables, influence the power to detect mediation on the $T_1 - T_2$ edge. We base our simula-
 383 tions on 192 Whole Blood trios that were positive for mediation under the GMAC mediation
 384 test. We consider four simulation scenarios: (i) when the inference model includes a few false
 385 positive confounders (denoted LTMS) (ii) when the inference model is missing a few true con-
 386 founders with weak effects (denoted STMS) (iii) when the inference model is correct (denoted

CMS) (iv) when the inference model is missing many true confounders with moderate to strong effects (SIMS)(See Supplementary Text 1 for details and methods).

Our simulations show that when the inference model contains most of the influential confounding variables, the correct inference for the $T_1 - T_2$ edge can typically be recovered (Table 4). The first two simulation scenarios represent relatively minor changes in the confounding variables used in the inference model, either by omitting or including weak confounders. In these scenarios, correct inference could typically be achieved, with the exception of a small subset of trios where even the correct model failed to yield accurate results.

Table 4: Type II error rates for MRGN under model misspecification. This table reports the type I and type II error rates under four simulation scenarios investigating model misspecification. Small True Model Simulation (STMS): The trans gene is simulated using only the most significant confounders, while the inference model includes additional, less significant confounders, representing model overspecification. Large True Model Simulation (LTMS): The trans gene is simulated with a larger set of confounders, but the inference model omits weaker confounders, representing model underspecification. Correct Model Simulation (CMS) - The same model is used for both simulation and inference, assessing the stability of the permutation test when the model is correctly specified. Small Inference Model Simulation (SIMS) - The trans gene is generated using a full model, but inference is performed with only a small subset of top confounders.

Test Type	STMS	LTMS	CMS	SIMS
Parametric test (Type II error)	0.0729	0.0729	0.0729	0.2969

The final scenario (SIMS) reflects the most significant discrepancy between the true model that generates the data and model used to infer the causal model. Here, the inference model omitted an average of 28 confounding variables ($SD \approx 7$), many of which had moderate to strong confounding effects. In this scenario, the type II error rate climbed to as high as 30%, underscoring the critical need to retain high-impact confounders for accurate causal inference.

4 Discussion

In summary, we have developed MRGN to improve the power and accuracy of detecting key causal relationships in diverse genomic trios, particularly in the presence of multiple confounding variables. Our method adopts an extended interpretation of the PMR [7] and leverages conditional and marginal hypothesis testing to identify five basic causal trio structures. By coupling MRGN with a relaxed version of the adaptive confounder selection procedure described by [15], our method achieves efficient and precise inference of causal edges without the computational penalty typically associated with high-dimensional confounding.

Our simulations demonstrate that MRGN outperforms comparable methods in detecting key causal edges ($T_1 - T_2$) and mediation models (M1) under realistic settings. A major advantage of our approach is the avoidance of the multiple testing burden; by adjusting for selected confounders within a regression framework, MRGN incorporates numerous confounders without the need to infer a graph larger than three nodes. Furthermore, by integrating a permutation test for the $T_1 - T_2$ edge, our method achieves results comparable to GMAC. Our comparisons suggest that the permutation test offers a conservative measure, reducing the risk of Type I errors in the presence of non-normal data or rare genetic effects.

Our results also underscore the critical importance of accurate confounder selection in causal inference. Omitting key confounders drastically reduces the power to detect mediation, leading to Type II errors and the loss of regulatory insights. Conversely, the inclusion of false-positive confounders has a negligible impact, particularly when their effects are weak. This suggests that in practical applications, a liberal strategy favoring the inclusion of many potential confounders is preferable. Ultimately, MRGN balances this requirement without excessive computational cost, making it a robust tool for genomic research.

PC-based methods such as MRPC initialize learning by filtering edges from a fully connected graph structure. After removing edges with weak marginal relationships, the remaining

edges are tested sequentially by conditioning on an increasing neighborhood of nodes [7, 29]. Because the number of tests required to determine the existence of an edge is unknown a priori, MRPC utilizes sequential FDR control [30]. Consequently, the testing burden is heavily dependent on graph connectivity, signal strength, and the total number of nodes.

Critically, MRPC treats confounding variables as additional nodes within the graph. When confounders are included, the number of conditional tests increases significantly, and only edges with strong supporting evidence survive. Thus, MRPC inherently infers a larger network than MRGN for the same biological question. Additionally, the adaptive FDR can result in significance thresholds that deviate substantially from the fixed $\alpha = 0.01$ employed by MRGN. As demonstrated by our simulations, this causes MRPC to act more conservatively in the presence of confounding variables.

While our simulations and real data analysis demonstrate the flexibility of MRGN in robustly estimating causal relationships, the method is currently limited to trios containing a genetic variant that acts as an instrumental variable. Similar to MRPC, MRGN relaxes strict instrumental variable assumptions by empirically validating the association between the variant and the phenotype(s). We are currently extending this approach to develop a general MRGN algorithm capable of accommodating larger networks and addressing other classes of confounding variables.

Data access

Data from The Genotype-Tissue Expression (GTEx) Project version 8 were obtained from dbGaP (<http://www.ncbi.nlm.nih.gov/gap>) through accession number phs000424.v8.pht002741.v8. Open-source R packages used in this manuscript include MRGN (version 1.3.7.3); <https://github.com/audreyfulab/MRGN>, MRPC (version 3.1.0); <https://cran.r-project.org/web/packages/MRPC/index.html>, and GMAC (version 3.1); <https://cran.r-project.org/web/packages/GMAC/index.html>. R code for simulations and applications is available at ???. Detailed results generated by our analyses are included in the supplementary material.

Competing interest statement

The authors declare no competing interests.

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Author Contributions

J.K and A.Q.F designed the study, J.K developed the tools and analyzed the data, and J.K and A.Q.F wrote the manuscript.

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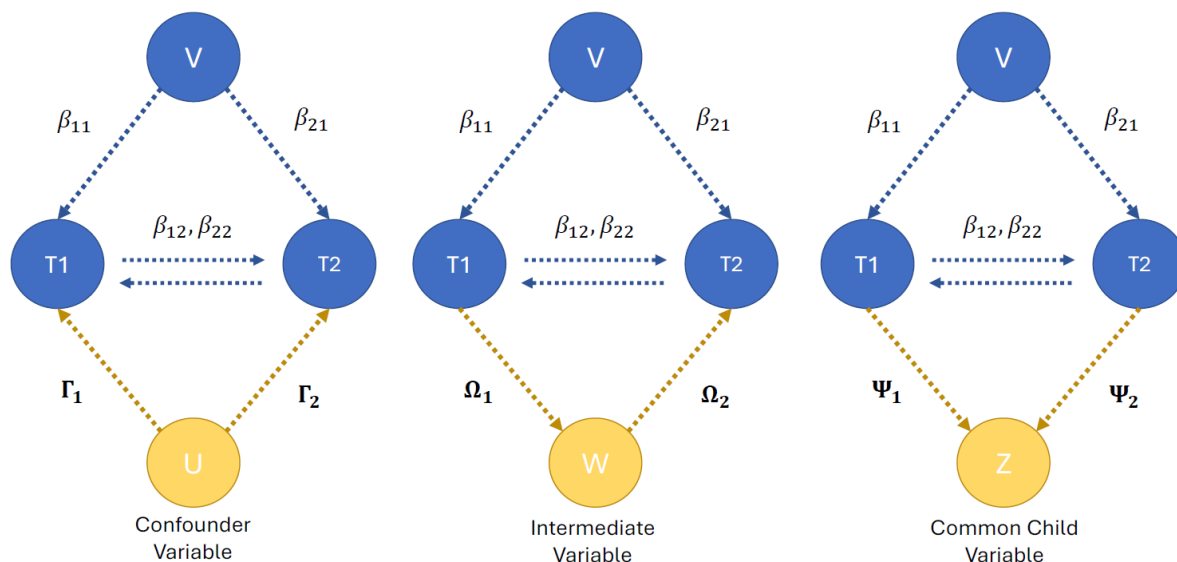
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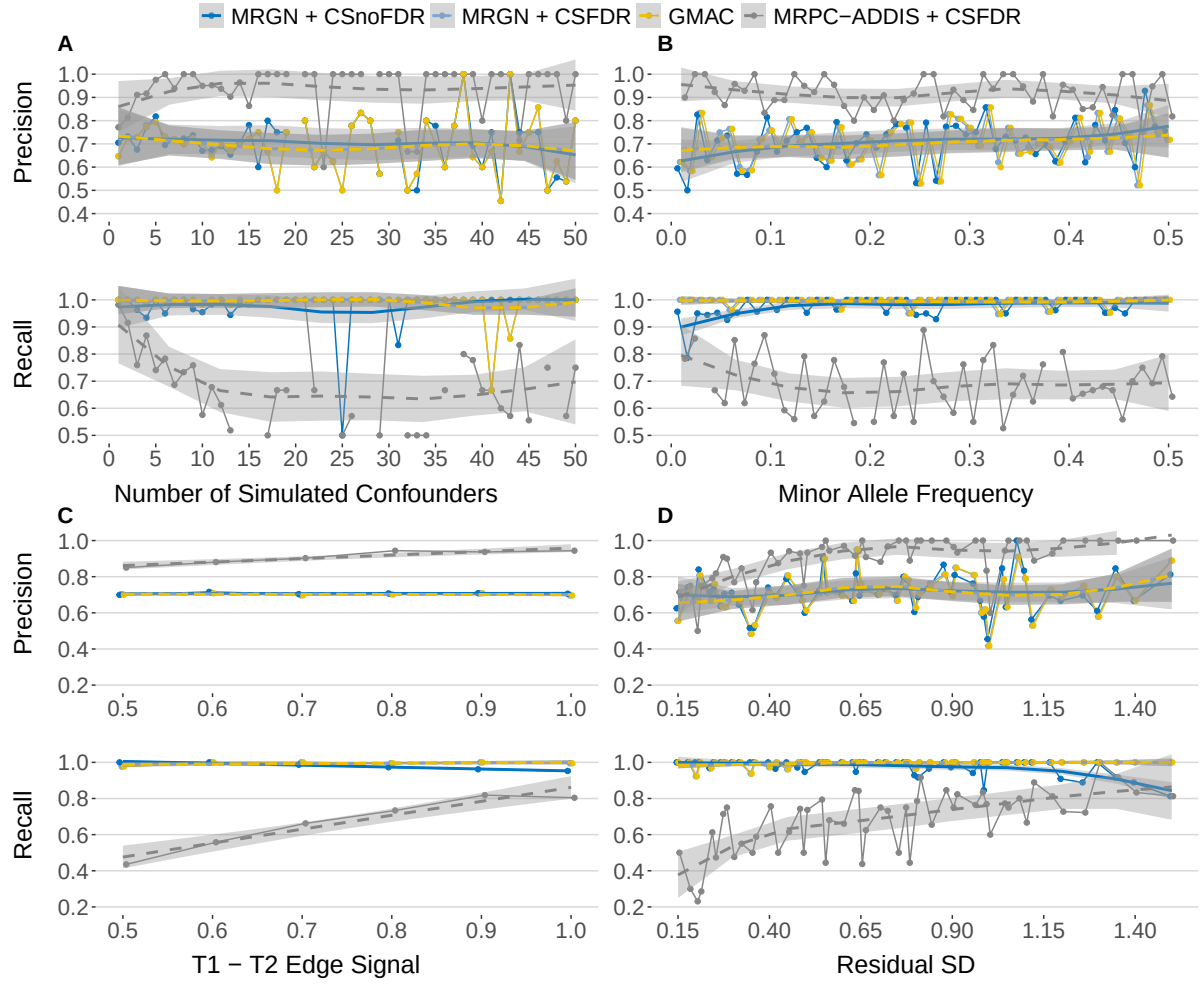
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559 **Tables**

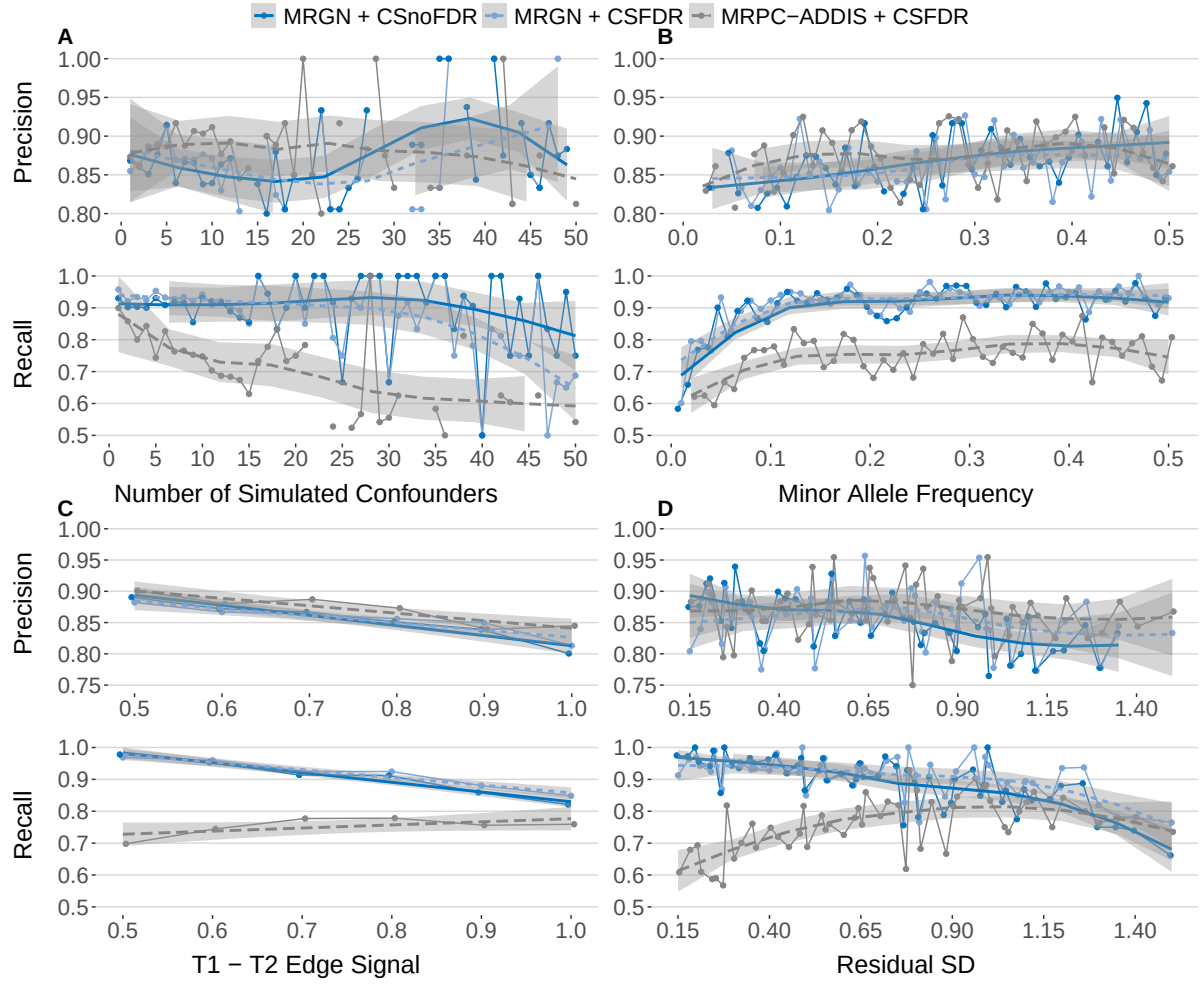
560 **Supplementary Material**



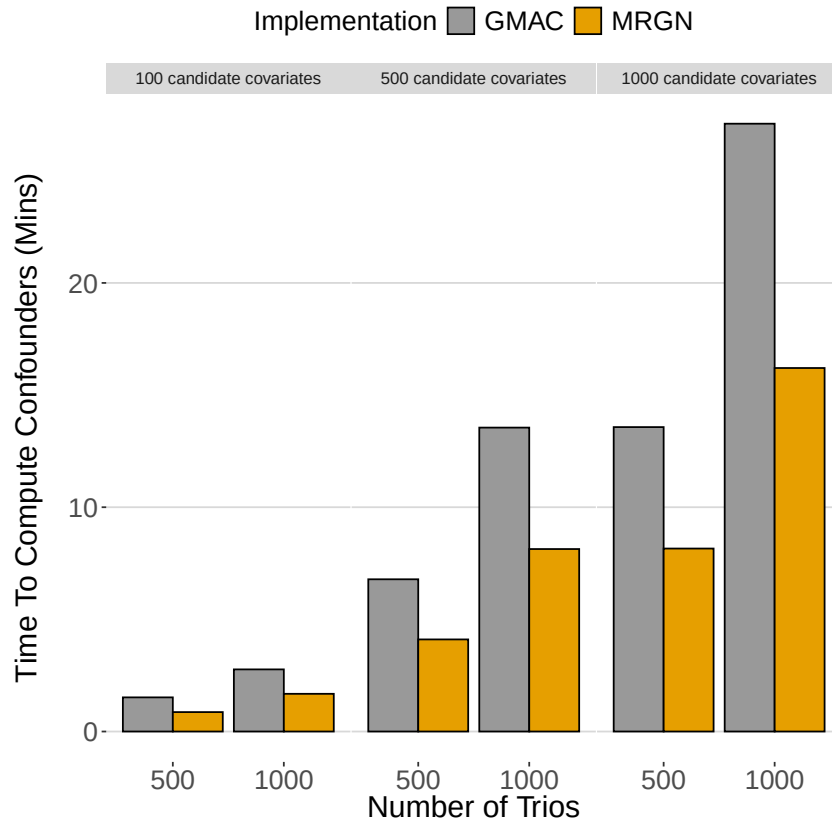
Supplementary Figure 1: The three types of confounding variables to consider in causal inference of genetic trios. A confounder (left) is a gene which regulates both molecular phenotypes, an intermediate variable (center) is a gene which mediates part of the effect of the genetic variant and its target, and a common child (right) is a shared target of both molecular phenotypes.



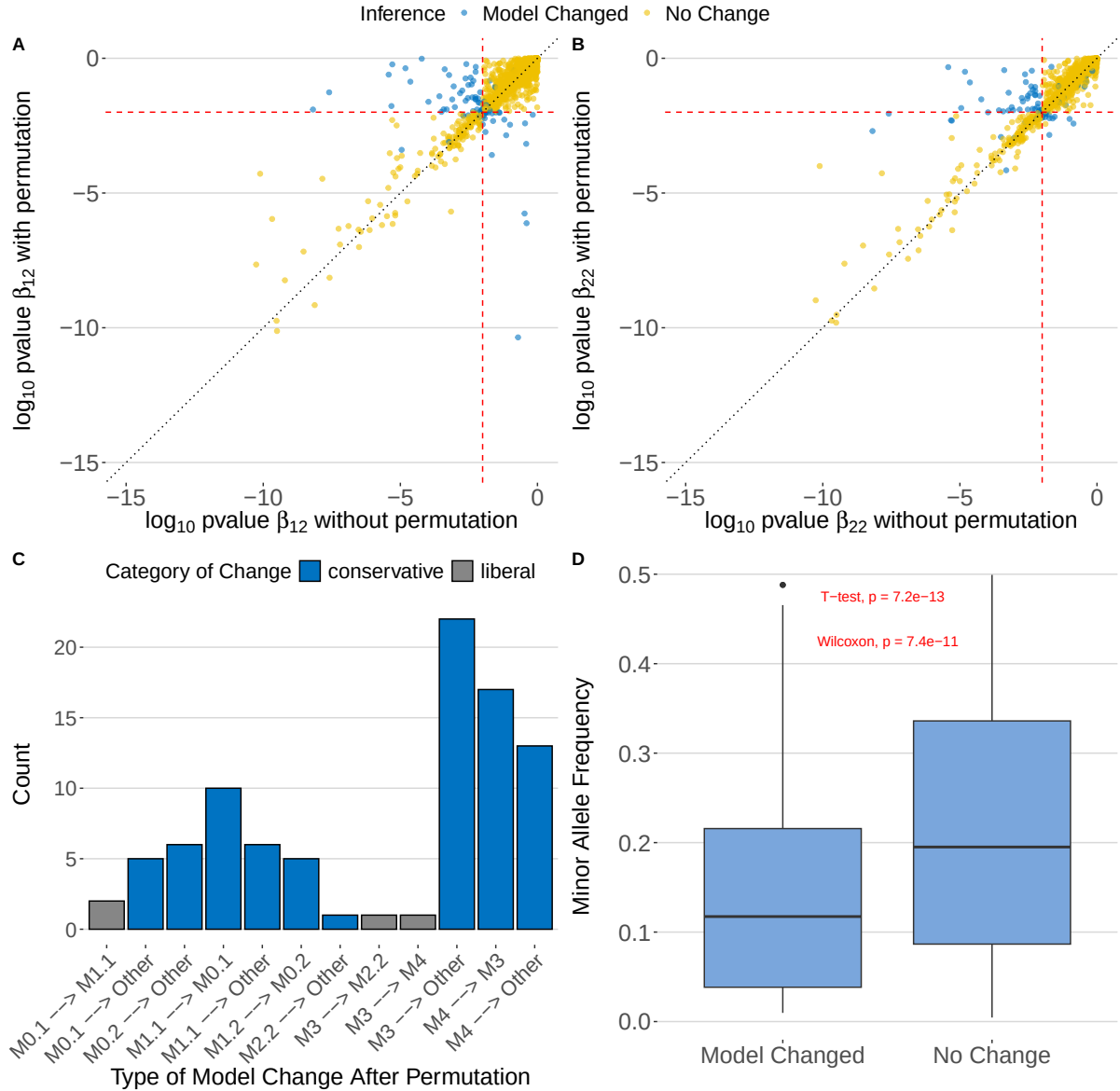
Supplementary Figure 2: Precision-recall curves for predicting the $T_1 - T_2$ edge across multiple simulation parameters. (A) the number of simulated hidden confounders (B) Minor allele frequency of the genetic variant, (C) The signal strength of the $T_1 - T_2$ edge, and (D) The standard deviation of the regression residuals.



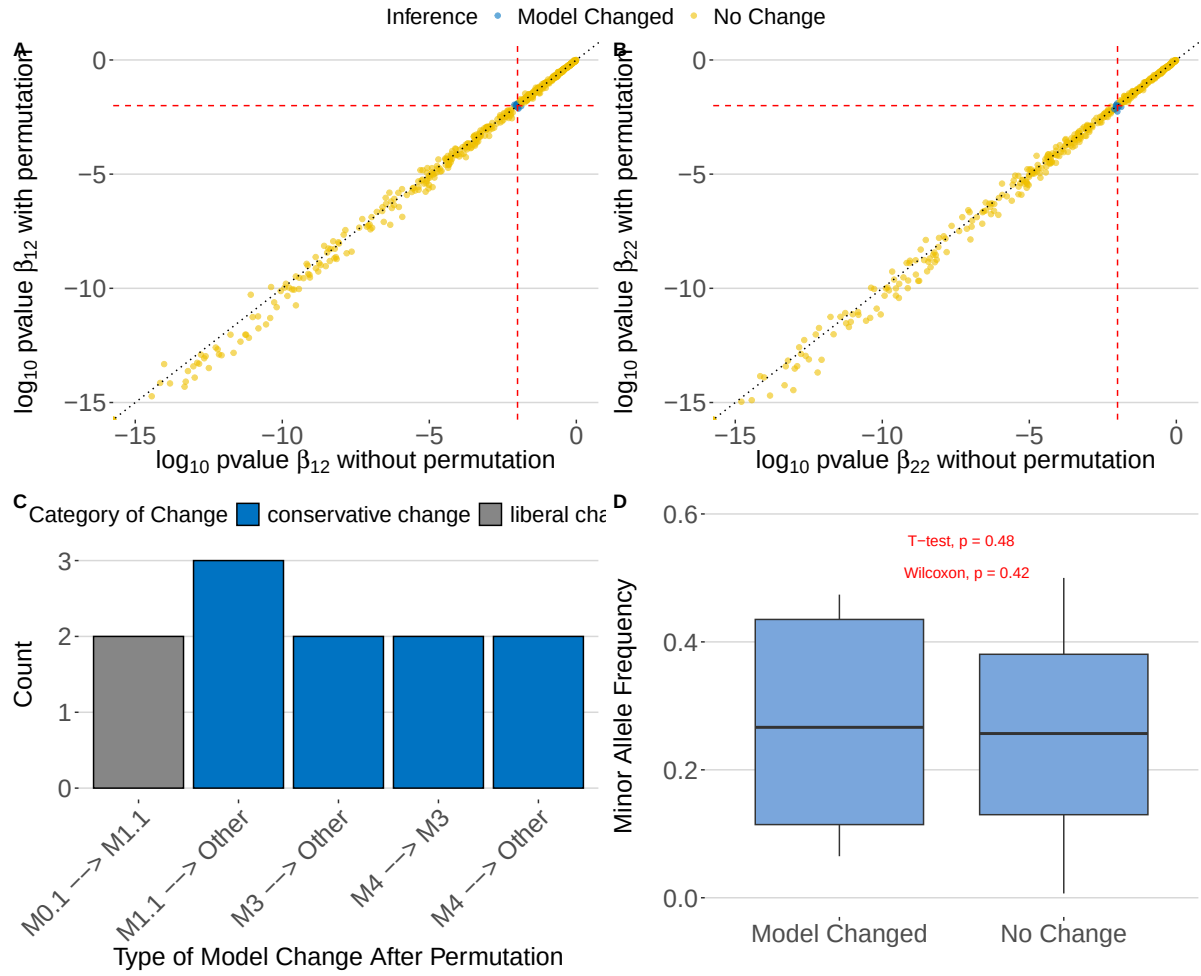
Supplementary Figure 3: Precision and recall curves of edge-based prediction performance for MRGN, and MRPC with confounder selection. (A) the number of simulated hidden confounders (B) Minor allele frequency of the genetic variant, (C) The signal strength of the $T_1 - T_2$ edge, and (D) The standard deviation of the regression residuals.



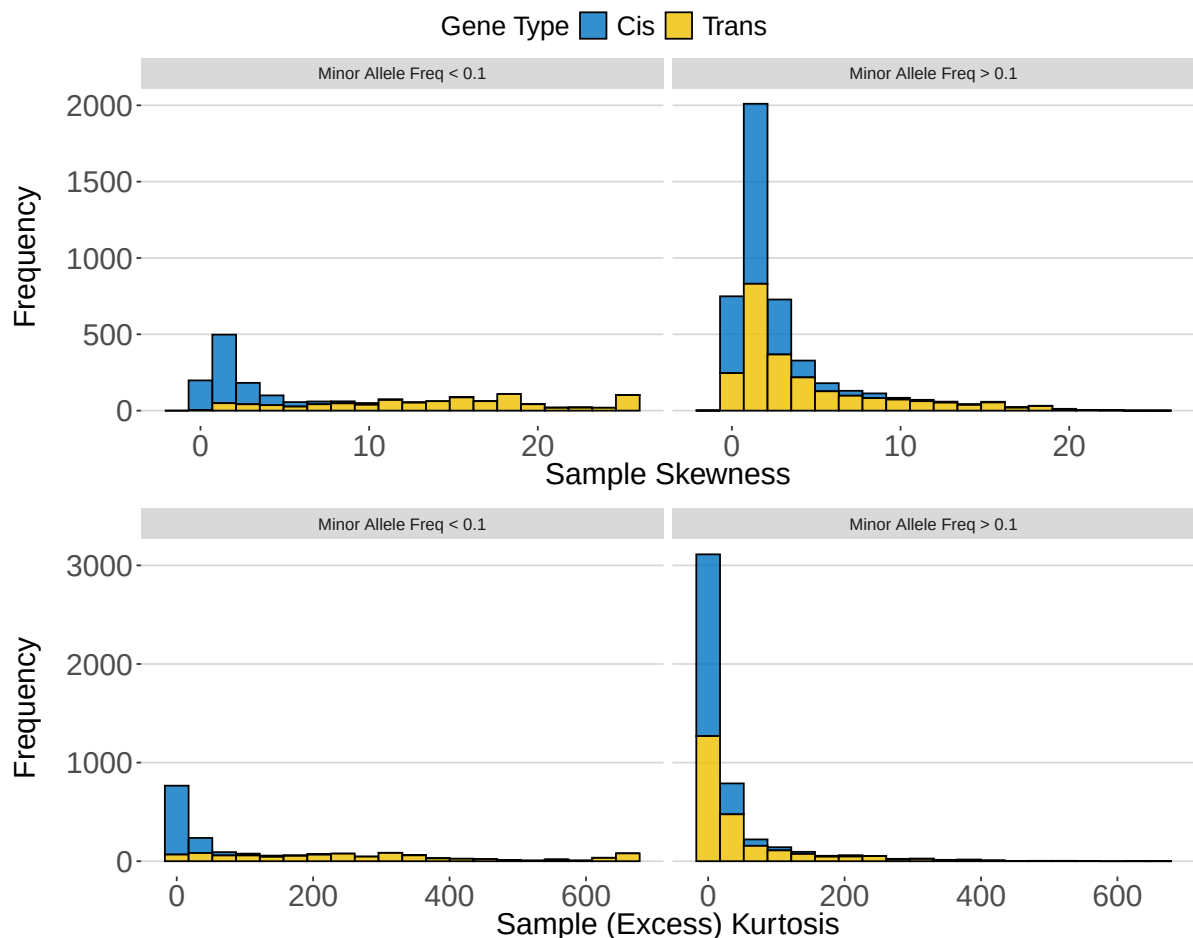
Supplementary Figure 4: Comparison of computation times between our implementation of the CSFDR method for confounder selection and that of the GMAC method [15]. Computation times are shown across different numbers of input trios and sizes of the candidate confounder pool.



Supplementary Figure 5: Impact of the causal permutation test on inference and categorization of model changes for GTEx whole blood trios. (A) A p -value plot (log₁₀ scale) comparing the observed p -value from the Wald test for $\beta_{12} : T_1 \perp\!\!\!\perp T_2 \mid (V, U)$ with the p -value from the permutation test. (B) A p -value plot for the conditional test $\beta_{22} : T_2 \perp\!\!\!\perp T_1 \mid (V, U)$ showing the permutation-based p -values (y-axis) versus the standard Wald test. Red dotted lines indicate the significance threshold on the log scale. (C) Categorization and counts of inferred model changes resulting from using the permutation test instead of the standard Wald test. (D) Minor allele frequencies of trios whose inferred models changed compared with those that did not change after applying the permutation test for β_{12} and β_{22} .

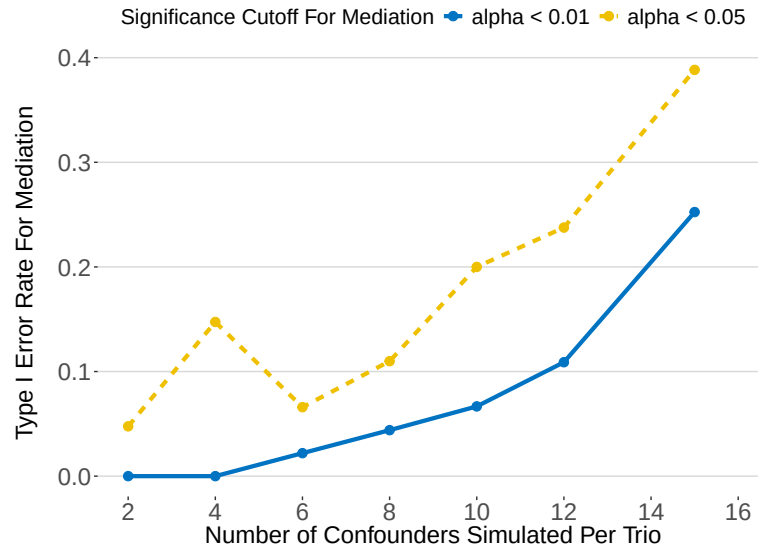


Supplementary Figure 6: Impact of the permutation test on inference and categorization of inferred model changes for simulated trios. (A) p -value plot ($\log_{10}$ scale) comparing the observed p -value from the Wald test for $\beta_{12} : T_1 \perp\!\!\!\perp T_2 \mid (V, U)$ with the p -value from the permutation test. (B) Same as (A) but for the conditional test $\beta_{22} : T_2 \perp\!\!\!\perp T_1 \mid (V, U)$. (C) Categorization and counts of inferred model changes resulting from using the permutation test instead of the standard Wald test. (D) Distribution of allele frequencies for trios whose inferred models changed using the permutation test compared with trios with no model change.



Supplementary Figure 7: Skewness and kurtosis for cis and trans genes by allele frequency. Stacked histograms showing the sampling distributions of sample skewness and excess kurtosis for cis and trans genes in GTEx trios. Results are segregated by the minor allele frequency of the genetic variant, highlighting differences between common and rare alleles.

Supplementary Figure 8: Example of a trio with a rare variant acting as an outlier. In this trio, a single individual homozygous for the minor allele drives non-normality in trans-gene expression relative to the population. This individual (yellow point) also shows elevated expression of the cis gene. Initially, MRGNTrio inferred model M4 for this trio, while GMAC inferred no mediation. Applying the permutation test caused MRGNTrio to recover the same inference (the null model) as GMAC.



Supplementary Figure 9: Results of simulations investigating type I error rate (false mediation) under the null model. Trios were simulated under the null model for each of seven scenarios: 2, 4, 6, 8, 10, 12 or 15 confounders, excluding common child and intermediate variables. The GMAC method was applied to trios in each scenario using CSFDR confounder selection.

562 Supplementary tables

Supplementary Table 1: A complete list of simulated trios, their simulation parameters, and inference results under MRGNTrio, MRPC-ADDIS, and GMAC.

Supplementary Table 2: All analyzed GTEx whole-blood trios, including genetic annotations (locus, variant ID, gene name, etc.) and inference results from MRGNTrio, MRPC-ADDIS, and GMAC.

Supplementary Table 3: This table compares MRGN, MRPC, and GMAC on results from application to GTEx whole blood tissue. Results compared between MRGN with CSnoFDR confounder selection to MRPC and GMAC.

Supplementary Text 1: Detailed supplementary text to explain the four simulation scenarios presented under the section “MRGN is robust to model misspecification in causal mediation”. This text also includes supplementary examples with output illustrating differences in causal inference between MRGN, GMAC, and MRPC on selected GTEx and simulated trios.